

1. Document Information

Field	Value
Project Name	Retail Sales Performance Analysis
Document	Data Model & DAX Documentation
Version	2.0
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Tools	PostgreSQL, Power BI, DAX
Date	09-Aug-2026
Status	Completed

2. Data Model Overview

The Power BI model uses a combination of fact, dimension, and bridge tables to organize the retail data for analysis.

The model is designed to support sales, customer, product, and campaign analysis while keeping the relationships between the tables clear.

3. Table Roles

Table	Type	Purpose
fact_transaction	Fact	Stores transaction-level sales data
dim_customer	Dimension	Stores customer attributes
dim_product	Dimension	Stores product attributes
dim_campaign	Dimension	Stores campaign information
bridge_campaign_customer	Bridge	Links customers with campaigns
fact_coupon_redemption	Fact	Stores coupon redemption activity

4. Relationships

The main relationships in the model connect transactions with customers and products, and campaign data with targeted customers and redemptions.

Key relationships include:

- dim_customer → fact_transaction
- dim_product → fact_transaction
- dim_campaign → bridge_campaign_customer
- dim_campaign → fact_coupon_redemption
- bridge_campaign_customer → fact_coupon_redemption where applicable

The model uses the appropriate relationship direction to allow dimensions to filter the related fact data.

5. Table Grain

Table	Grain
fact_transaction	One product transaction within a basket
dim_customer	One customer/household
dim_product	One product
dim_campaign	One campaign
bridge_campaign_customer	One customer-campaign relationship
fact_coupon_redemption	One coupon redemption record

6. DAX Measures

Sales KPIs

1. Total Revenue = SUM(fact_transaction[sales_value])
2. Total Orders = DISTINCTCOUNT(fact_transaction[basket_id])
3. Total Customers = DISTINCTCOUNT(fact_transaction[household_key])
4. Average Order Value = DIVIDE([Total Revenue], [Total Orders])

Customer KPIs

1. Repeat Customers = COUNTROWS(FILTER(VALUES(fact_transaction[household_key]), CALCULATE (DISTINCTCOUNT(fact_transaction[basket_id])) > 1))
2. Repeat Customer Rate = DIVIDE([Repeat Customers], [Total Customers])
3. Average Basket Size = DIVIDE(SUM(fact_transaction[quantity]), DISTINCTCOUNT(fact_transaction[basket_id]))

Campaign KPIs

1. Targeted Customers = DISTINCTCOUNT(bridge_campaign_customer[household_key])
2. Redeemed Customers = DISTINCTCOUNT(fact_coupon_redemption[household_key])
3. Redemption Rate = DIVIDE([Redeemed Customers], [Targeted Customers])

7. Modeling Considerations

The following points were considered while building the model:

- Fact tables are used for transactional and activity-level data.
- Dimension tables provide descriptive attributes for filtering and analysis.
- The bridge table handles the customer-to-campaign relationship.
- Unique customer and basket counts are calculated using DISTINCTCOUNT.
- Division-based KPIs use DIVIDE to handle zero denominators safely.
- The model was designed to support interactive Power BI filtering.

8. Validation

Key KPIs and calculations were cross-checked against the SQL analysis to ensure consistency between the data model and dashboard results.

9. Conclusion

The data model provides a structured foundation for analyzing sales, customers, products, and campaigns.

The DAX measures convert the underlying data into business-focused KPIs used throughout the Power BI dashboard.